COURSE 6 · BIOLOGY

Biology

Theory reader, audio edition

The chemistry of life, cells, metabolism, DNA and genes, inheritance, evolution, the human body, microbes and immunity, and ecosystems.

9 modules · equations spoken in words; numbered equations appear in the Visual Companion.

Before you start

How to use this reader

This is the audio edition. It is written to be listened to in a text-to-speech app as comfortably as read on the page. Equations are spoken in words, and each important one carries a number like this: [Equation 6.1-1]. The same number appears in the Visual Companion, where the equation is printed in ordinary notation.

Each module takes about twenty to thirty minutes to listen to. A good weekly rhythm at four hours: listen on the first day, look over the Visual Companion with a pencil, try the explorer, then spend the remaining time in the workbook.

The level sits between final-year high school and first-year university. Nothing is assumed beyond what you met at school, and it is fine if that was a long time ago. Where a first-year idea appears, it is introduced from the ground up.

In this course

6.1 The chemistry of life
6.2 Cells
6.3 Metabolism
6.4 DNA, genes and protein synthesis
6.5 Inheritance and genetics
6.6 Evolution and natural selection
6.7 Human body systems
6.8 Microbes, immunity and disease
6.9 Ecology and ecosystems

Biology is chemistry and physics organised into living systems. This course runs from molecules to ecosystems: how cells work, how they get energy, how information is stored and copied in DNA, how traits are inherited and how populations evolve, then up to bodies, disease and the living planet. It builds directly on the organic chemistry of Module 4.8 and the exponential growth of Module 2.7.

Module 6.1

The chemistry of life

Living things are made of the same atoms as rocks, organised into four big families of molecules. Understanding water, those molecules and the enzymes that drive their reactions is the base layer of biology.

By the end of this module you can

- Explain why water's properties matter to life
- Describe carbohydrates, lipids, proteins and nucleic acids
- Explain how enzymes work and what affects them

Water

Cells are about seventy per cent water. Water is polar, as Module 4.3 explained, and hydrogen bonds between its molecules give it unusual properties: it dissolves many substances, so it is the medium for the reactions of life; it has a high specific heat, so it buffers temperature; and ice floats, insulating lakes so life survives beneath. Few other molecules could do all this.

Four families of biological molecules

Carbohydrates, such as glucose, starch and cellulose, are fuels and building materials. Lipids, including fats and oils, store energy compactly and form the membranes around every cell, because one end of the molecule likes water and the other avoids it. Proteins are chains of twenty kinds of amino acids, folded into precise three-dimensional shapes; they do most of the work of the cell, as enzymes, transporters, signals and structures. Nucleic acids, DNA and RNA, store and transmit genetic information.

Most of these are polymers built by condensation reactions, which release water as units join [Equation 6.1-1]. Digestion reverses this: hydrolysis uses water to break the polymers apart.

Enzymes

Enzymes are biological catalysts, mostly proteins. Each has an active site shaped to fit particular substrate molecules, an idea often pictured as a lock and key, though the fit tightens when the substrate binds. By holding substrates in the right orientation and straining their bonds, enzymes lower the activation energy from Module 4.6, speeding reactions millions of times.

Because their action depends on shape, enzymes are sensitive to conditions. Rate rises with temperature up to an optimum, around thirty-seven degrees for human enzymes, then falls sharply as the protein unfolds, or denatures. Each enzyme also has an optimum pH. Rate rises with substrate concentration until every active site is busy, then levels off [Equation 6.1-2].

Key ideas to carry forward

- Water's polarity and hydrogen bonding make it the medium of life
- Carbohydrates, lipids, proteins, nucleic acids
- Polymers form by condensation and break by hydrolysis
- Enzymes lower activation energy; they denature outside their optimum

Module 6.2

Cells

The cell is the smallest unit of life. Every organism, from a bacterium to a blue whale, is one cell or many, and every cell comes from a pre-existing cell.

By the end of this module you can

- State cell theory and compare prokaryotic and eukaryotic cells
- Describe the main organelles and the cell membrane
- Explain diffusion, osmosis and active transport, and why cells are small

Cell theory

Robert Hooke named cells in 1665 after looking at cork through a microscope. By the mid-1800s, cell theory was established: all living things are made of cells, the cell is the basic unit of life, and cells arise only from other cells. That last point ended the old idea of spontaneous generation.

Two kinds of cell

Prokaryotic cells, bacteria and archaea, are small and simple, with DNA floating in the cytoplasm and no nucleus. Eukaryotic cells, which make up animals, plants, fungi and protists, are larger and divided into compartments called organelles. The nucleus holds DNA. Mitochondria release energy from food. Ribosomes make proteins. The endoplasmic reticulum and Golgi apparatus fold, package and ship them. Plant cells add a rigid cell wall, a large vacuole, and chloroplasts for photosynthesis.

Mitochondria and chloroplasts have their own small loops of DNA and divide like bacteria. They are descended from bacteria that took up residence inside an ancestral cell more than a billion years ago, an idea championed by Lynn Margulis that was ridiculed before it was confirmed.

The membrane and transport

Every cell is wrapped in a membrane: a double layer of phospholipids studded with proteins. It lets some substances through and blocks others. Small molecules like oxygen diffuse across, moving from high to low concentration. Water moves by osmosis, towards the side with more dissolved solute. Cells also pump substances against their concentration gradient, using energy; this is active transport.

Cells are small for a reason you met in Module 1.3: the square-cube law. As a cell grows, its volume, and its needs, grow faster than its surface, through which everything must pass [Equation 6.2-1]. Beyond a certain size, the surface cannot keep up. Big organisms are made of many small cells, not a few giant ones.

Key ideas to carry forward

- All living things are made of cells; cells come from cells
- Prokaryotes: no nucleus. Eukaryotes: nucleus and organelles
- Diffusion and osmosis are passive; active transport uses energy
- Surface-to-volume ratio limits cell size

Module 6.3

Metabolism

Every living thing runs on a constant flow of energy. Plants capture it from sunlight; almost everything else gets it by breaking down food. The two processes, photosynthesis and respiration, are mirror images.

By the end of this module you can

- Describe ATP as the cell's energy currency
- Summarise photosynthesis and aerobic respiration
- Compare aerobic and anaerobic respiration

ATP

Cells cannot use glucose directly to power their machinery. Instead they transfer its energy, in small steps, into a molecule called ATP, adenosine triphosphate. When ATP loses a phosphate group to become ADP, it releases a convenient packet of energy [Equation 6.3-1]. Muscles contracting, nerves firing and proteins being built all spend ATP. You recycle roughly your own body weight of it every day.

Photosynthesis

In the chloroplasts of plants and algae, the green pigment chlorophyll absorbs light, mostly red and blue, and uses its energy to split water, releasing oxygen. The energy captured is then used to build glucose from carbon dioxide [Equation 6.3-2]. Almost all the oxygen in the atmosphere, and almost all the energy in food chains, comes from photosynthesis. In chemistry terms, carbon dioxide is reduced, the redox idea from Module 4.7.

Respiration

Aerobic respiration runs the other way, oxidising glucose back to carbon dioxide and water [Equation 6.3-3]. It happens in stages: glycolysis in the cytoplasm splits glucose; then, in the mitochondria, the Krebs cycle and the electron transport chain extract the rest of the energy. One glucose molecule yields around thirty ATP. Oxygen is the final electron acceptor, which is why you need to breathe.

Without oxygen, cells fall back on anaerobic respiration, which stops after glycolysis and yields only two ATP per glucose. Your muscles make lactate this way during a sprint; yeast makes ethanol and carbon dioxide, which gives us bread and beer [Equation 6.3-4].

Key ideas to carry forward

- ATP is the energy currency: ATP → ADP + P releases energy
- Photosynthesis: $CO_2 + H_2O$ + light → glucose + O_2
- Respiration: glucose + O_2 → $CO_2 + H_2O$ + about 30 ATP
- Anaerobic: only 2 ATP; lactate in muscle, ethanol in yeast

Module 6.4

DNA, genes and protein synthesis

DNA is the instruction manual of life, written in a four-letter alphabet. This module explains how the information is stored, how it is copied, and how it is read to build proteins.

By the end of this module you can

- Describe the structure of DNA and base pairing
- Explain DNA replication
- Describe transcription, translation and the genetic code, and what mutations do

The double helix

DNA is a polymer of nucleotides, each made of a sugar, a phosphate and one of four bases: adenine, thymine, guanine and cytosine. Two strands twist around each other in a double helix, held together by hydrogen bonds between bases. Adenine always pairs with thymine, and guanine with cytosine [Equation 6.4-1]. The structure was published in 1953 by James Watson and Francis Crick, relying crucially on X-ray images taken by Rosalind Franklin.

Copying DNA

Because each base has only one partner, each strand carries the information to rebuild the other. Before a cell divides, the helix unzips and enzymes build a new partner strand for each old one. Each new DNA molecule keeps one old strand and gains one new one, so replication is called semi-conservative. Errors are rare, about one in a billion bases after proofreading.

From gene to protein

A gene is a stretch of DNA that codes for a protein. In transcription, the gene is copied into a messenger RNA molecule, which leaves the nucleus. In translation, ribosomes read the RNA three bases at a time. Each triplet, called a codon, specifies one amino acid, and transfer RNA molecules bring the right amino acids in order [Equation 6.4-2]. With four bases taken three at a time there are sixty-four codons for twenty amino acids, so the code has redundancy [Equation 6.4-3]. The same code is used by almost every organism on Earth, strong evidence for common ancestry.

A mutation is a change in the DNA sequence. Many have no effect, especially if they swap one codon for another meaning the same amino acid. Some change an amino acid and alter a protein's shape: sickle cell disease comes from a single base change in the haemoglobin gene. Mutations are also the raw material of evolution, Module 6.6.

Key ideas to carry forward

- DNA: double helix, A–T and G–C pairing
- Replication is semi-conservative
- DNA → mRNA (transcription) → protein (translation)
- Codons are triplets; the code is universal and redundant

Module 6.5

Inheritance and genetics

Why do children resemble their parents, but not exactly? Gregor Mendel's pea experiments in the 1860s revealed the rules, and they reduce to the probability you met in Module 1.5.

By the end of this module you can

- Use genes, alleles, genotype and phenotype correctly
- Predict offspring ratios with Punnett squares, including sex linkage
- Explain meiosis and the sources of genetic variation

Genes and alleles

A gene can come in different versions called alleles. You carry two copies of most genes, one from each parent. If both alleles are the same you are homozygous; if different, heterozygous. Your genotype is the alleles you have; your phenotype is the trait you show. A dominant allele shows its effect even with one copy; a recessive allele only shows when both copies are recessive.

Punnett squares

Mendel crossed tall and short pea plants. A Punnett square lays out the possible combinations of parental alleles. Two heterozygous parents, A a crossed with A a, give one AA, two A a and one a a in every four offspring on average: a three to one ratio of dominant to recessive phenotype [Equation 6.5-1]. Each offspring is an independent draw, so these are probabilities, not guarantees.

Some traits don't follow simple dominance. In incomplete dominance the heterozygote is intermediate. Many traits, like height, are polygenic, shaped by many genes plus the environment, so they vary continuously and follow something like the normal distribution.

Sex chromosomes and meiosis

Humans have twenty-three pairs of chromosomes; one pair decides sex, XX or XY. Genes on the X chromosome show sex-linked inheritance: a recessive X-linked allele, like that for red-green colour blindness, shows up far more often in males because they have only one X.

Sex cells are made by meiosis, a cell division that halves the chromosome number so that fertilisation restores it. Meiosis shuffles chromosomes independently and swaps segments between partner chromosomes, called crossing over. With twenty-three pairs, independent assortment alone gives more than eight million combinations per parent [Equation 6.5-2], which is why siblings differ.

Bridge from the social sciences

Genetics turns on probability and populations, so it sits naturally alongside social statistics. Heritability, for example, is a population statistic about variance, not a statement about how much of one person's trait is 'genetic'.

Key ideas to carry forward

- Genotype: alleles you carry. Phenotype: the trait shown
- Aa × Aa gives 3:1 dominant:recessive phenotypes
- X-linked recessive traits are more common in males
- Meiosis halves chromosome number and creates variation

Module 6.6

Evolution and natural selection

Evolution is the organising idea of biology. Darwin and Wallace saw that variation, inheritance and differential survival together produce adaptation, and modern genetics has confirmed and extended their insight.

By the end of this module you can

- Explain natural selection and adaptation
- Describe the evidence for evolution
- Explain speciation and genetic drift, and correct common misconceptions

Natural selection

Charles Darwin and Alfred Russel Wallace independently arrived at the same idea, published together in 1858. Individuals in a population vary. Some of that variation is inherited. More offspring are born than can survive. Individuals whose inherited traits suit their environment are more likely to survive and reproduce, so those traits become more common in the next generation. Over many generations, this produces adaptation.

Selection acts on individuals, but evolution happens to populations: it is a change in allele frequencies over time [Equation 6.6-1]. A trait giving even a small advantage, a few per cent, can spread through a population in some hundreds of generations, as the Visual Companion shows.

The evidence

The fossil record shows life changing over hundreds of millions of years, with transitional forms such as feathered dinosaurs and early whales with legs. Comparative anatomy reveals homologous structures: the same bones in a human arm, a bat's wing and a whale's flipper. DNA sequences give a family tree that matches the one built from anatomy, and the near-universal genetic code points to common ancestry. And evolution is observed today, in antibiotic resistance, pesticide resistance and Darwin's finches, whose beak sizes changed measurably after droughts in the 1970s.

Drift, speciation and misconceptions

Not all change is adaptive. In small populations, chance alone can make alleles more or less common: genetic drift. New species form when populations become separated, by a mountain range or an ocean, and diverge until they can no longer interbreed.

Some misconceptions are worth clearing away. Individuals do not evolve during their lives; populations do. Organisms do not develop traits because they need them; variation arises randomly and selection sorts it. Evolution is not a ladder with humans at the top; it is a branching bush, and every living species is equally evolved for its own circumstances. And 'theory' in science means a well-tested explanation, not a guess.

Bridge from the social sciences

Evolutionary ideas have been misapplied in social theory, notably in Social Darwinism. Biologically, 'fitness' means only reproductive success in a particular environment; it carries no moral meaning.

Key ideas to carry forward

- Natural selection needs variation, inheritance and differential reproduction
- Evolution = change in allele frequencies in a population
- Evidence: fossils, anatomy, DNA, observed change
- Drift is chance; speciation follows separation

Module 6.7

Human body systems

Your body is a society of about thirty trillion cells organised into tissues, organs and systems. They work together to keep the conditions inside you remarkably constant despite a changing world outside.

By the end of this module you can

- Describe the circulatory, respiratory, digestive and nervous systems
- Explain homeostasis and negative feedback
- Describe hormones and the control of blood glucose and temperature

Levels of organisation

Cells form tissues, tissues form organs, and organs form systems. The circulatory system, heart, blood and vessels, transports oxygen, nutrients and wastes. The heart is a double pump: one side sends blood to the lungs, the other to the rest of the body. Cardiac output, the volume pumped per minute, is heart rate times stroke volume [Equation 6.7-1], about five litres a minute at rest.

The respiratory system brings oxygen in and takes carbon dioxide out. In the lungs, about three hundred million tiny air sacs called alveoli give a surface area roughly the size of a tennis court, the square-cube law again. The digestive system breaks food into molecules small enough to absorb, using enzymes from Module 6.1, with the small intestine's folds and villi providing a huge absorbing surface.

Nerves and hormones

The nervous system sends fast electrical signals along neurons, up to about a hundred metres per second, and passes them between neurons at synapses using chemical messengers. The endocrine system sends slower, longer-lasting signals as hormones in the blood. Adrenaline prepares you for action; insulin controls blood sugar; growth hormone and sex hormones shape development.

Homeostasis

Homeostasis is keeping the internal environment stable: temperature, blood glucose, water and salt balance, pH. Most of it works by negative feedback. A change is detected, a response pushes it back, and the response switches off as conditions return to normal, like a thermostat.

After a meal, blood glucose rises; the pancreas releases insulin, which makes cells take up glucose and the liver store it as glycogen, so glucose falls back. Between meals, glucagon does the reverse. In type one diabetes the pancreas makes no insulin; in type two, cells respond poorly to it. Temperature works the same way: too hot and you sweat and flush; too cold and you shiver and your skin vessels narrow.

Key ideas to carry forward

- Cells → tissues → organs → systems
- Heart: double pump. Lungs: alveoli give huge surface area
- Nerves: fast, electrical. Hormones: slower, in the blood
- Homeostasis by negative feedback, e.g. insulin and glucagon

Module 6.8

Microbes, immunity and disease

Microbes are the most abundant life on Earth, and most are harmless or helpful. A few cause disease. Your immune system, and medicine's vaccines and antibiotics, are how we fight back.

By the end of this module you can

- Compare bacteria, viruses, fungi and protists
- Describe innate and adaptive immunity and how vaccines work
- Explain antibiotics, resistance and the basics of epidemic spread

The microbial world

Bacteria are single prokaryotic cells. Under good conditions some divide every twenty minutes, so one cell could in principle become billions in a day, the exponential growth of Module 2.7 [Equation 6.8-1]. Your gut holds trillions of them, helping digestion and training your immune system. Viruses are not cells at all: they are genetic material in a protein coat that can only reproduce by hijacking a host cell. Fungi include yeasts and moulds; protists include the parasite that causes malaria.

The germ theory of disease, established by Louis Pasteur and Robert Koch in the late 1800s, replaced older ideas like 'bad air'. Ignaz Semmelweis had earlier shown that handwashing cut deaths in childbirth dramatically, though he was ignored at the time.

Defences

Your first defences are barriers: skin, mucus, stomach acid. Behind them is the innate immune system, which responds quickly and generally, with inflammation and white blood cells that engulf invaders. The adaptive immune system is slower to start but highly specific. B cells make antibodies that lock onto particular molecules, called antigens, on a pathogen; T cells kill infected cells and coordinate the response.

Crucially, adaptive immunity remembers. After an infection, memory cells persist, so a second encounter brings a faster, bigger response, and often you never feel ill. Vaccines exploit this, training the immune system with a harmless version or piece of the pathogen. Edward Jenner's smallpox vaccine of 1796 began the story; smallpox was declared eradicated in 1980.

Antibiotics and epidemics

Antibiotics kill bacteria or stop their growth by targeting structures that human cells lack, such as bacterial cell walls. They do nothing against viruses. Overuse drives antibiotic resistance by natural selection, one of the most serious threats in modern medicine.

In an epidemic, the basic reproduction number R nought is the average number of new infections caused by one infected person in a fully susceptible population. If it is above one, cases grow.

Vaccination slows spread by reducing susceptible people; if enough are immune, the effective reproduction number falls below one and the disease declines even among the unvaccinated. This herd immunity threshold is one minus one over R nought [Equation 6.8-2].

Bridge from the social sciences

Epidemic models are a meeting point of biology and social science: contact patterns, behaviour and trust in health systems shape R as much as biology does.

Key ideas to carry forward

- Bacteria are cells; viruses need host cells
- Innate immunity: fast, general. Adaptive: slow, specific, with memory
- Vaccines train immune memory without the disease
- Herd immunity threshold = $1 - 1/R_0$; antibiotics don't work on viruses

Module 6.9

Ecology and ecosystems

Ecology studies how organisms interact with each other and their environment. Energy flows through ecosystems, nutrients cycle round them, and populations rise and fall within limits, all of which now carry a human signature.

By the end of this module you can

- Describe food chains, trophic levels and energy flow
- Describe the carbon cycle and human impacts
- Model population growth with exponential and logistic curves

Energy flow

Sunlight enters an ecosystem through producers: plants, algae and some bacteria that photosynthesise. Herbivores eat producers, carnivores eat herbivores, and decomposers break down the dead. Each step is a trophic level. Only about ten per cent of the energy at one level passes to the next [Equation 6.9-1]; the rest is lost as heat from respiration or in material that isn't eaten. That is why food chains are short and why big predators are rare.

Nutrient cycles

Energy flows through, but matter cycles. In the carbon cycle, photosynthesis draws carbon dioxide from the air into living things; respiration and decomposition return it. Some carbon is locked away for millions of years in fossil fuels and rocks. Burning fossil fuels returns that ancient carbon quickly, raising atmospheric carbon dioxide from about two hundred and eighty parts per million before industrialisation to over four hundred and twenty today, trapping heat by the greenhouse effect. Nitrogen and water cycle too, and fertiliser use has doubled the flow of usable nitrogen on Earth.

Populations

With unlimited resources, a population grows exponentially. In reality, food, space and predators limit it. The logistic model captures this: growth is fast at first, slows as the population approaches the carrying capacity K, and levels off [Equation 6.9-2]. Predator and prey populations often cycle, the famous example being lynx and hare furs traded in Canada, with predator peaks lagging prey peaks.

Biodiversity

Biodiversity is the variety of life at every level: genes, species and ecosystems. It underpins services we depend on, such as pollination, clean water and fertile soil. Australia is one of the world's most biodiverse countries, with most of its mammals, reptiles and frogs found nowhere else, and it also has one of the worst records of mammal extinction since European settlement. Habitat loss, invasive species, climate change and pollution are the main drivers.

Bridge from the social sciences

The logistic curve also models technology adoption and diffusion of innovations, and carrying capacity has a long and contested history in demography, from Malthus onward.

Key ideas to carry forward

- About 10% of energy passes up each trophic level
- Energy flows; matter cycles
- Logistic growth levels off at carrying capacity K
- Human activity is reshaping the carbon cycle and biodiversity